INCH-POUND

MIL-PRF-19500/678
1 July 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON,DUAL SCHOTTKY CENTER TAP
POWER RECTIFIER, SURFACE MOUNTED,
TYPES 1N6840U3 AND 1N6841U3, JAN, JANTX, JANTXV, JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, dual Schottky power rectifier in a surface mount package. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (U3, similar to SMD.5).

1.3 Maximum ratings.

Type	V_{RWM}	I_O (1) T_C= +100 °C	I_{FSM} T_C =+25 °C t_p= 8.3 ms	C_J at 5 V	T_{STG} and T_J
	V	A dc	A (pk)	pF	°C
1N6840	35	10	200	400	-65 to +150
1N6841	45	10	200	400	-65 to +150

(1) Derate linearly at 200 mA/°C from $T_J = T_C$ =+100°C to + 150°C.
(2) Each individual diode.
(3) Total package limited to 20 A.

1.4 Primary electrical characteristics. $R_{\theta JC}$ = 2.8°C/W maximum on each leg.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961
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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.X and 6.X).

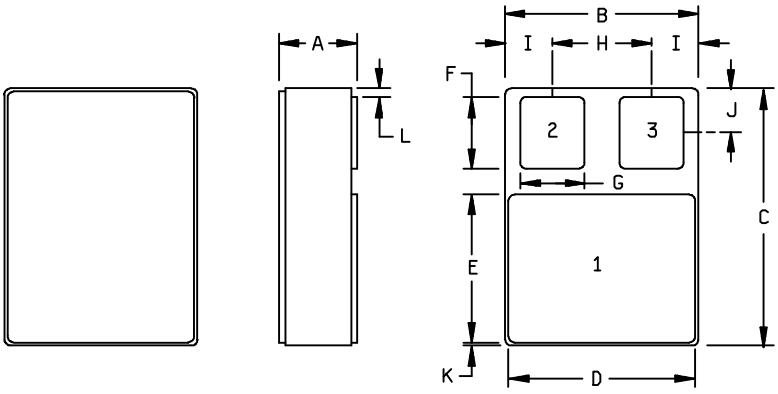
3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein. The US government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.4.1 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	0.111	0.122	2.82	3.10
B	0.291	0.301	7.39	7.65
C	0.395	0.405	10.03	10.29
D	0.281	0.291	7.14	7.39
E	0.220	0.230	5.59	5.84
F	0.115	0.125	2.92	3.18
G	0.090	0.100	2.29	2.54
H	0.125	0.135	3.68	3.94
I	0.073 TYP.	-	1.85 TYP.	-
J	0.083 TYP.	-	2.11 TYP.	-
K	0.005 TYP.	-	0.13 TYP.	
L	0.015 TYP.	-	0.38 TYP.	-

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal 1 is common cathode.
4. Terminal 2 is anode 1.
5. Terminal 3 is anode 2.

SCHEMATIC

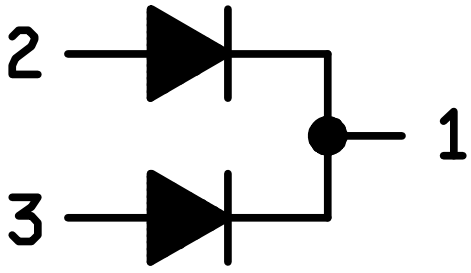


FIGURE 1. Physical dimensions and configuration (U3, similar to SMD.5).

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3.6 Electrical performance characteristics Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I, group A herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.3 Screening (JANTX, JANTXV and JANS levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c (1)	Method 3101 (see 4.3.2).	Method 3101 (see 4.3.2).
9 and 10	Not applicable	Not applicable
11	V_{F2} and I_{R1}	V_{F2} and I_{R1}
12	Condition A. See 4.3.1, t = 240 hours.	Condition A. See 4.3.1, t = 48 hours.
13	Subgroups 2 and 3 of table I herein; ΔV_{F2} = ± 50 mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.	Subgroup 2 of table I herein; ΔV_{F2} = ± 50 mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.

(1) Shall be performed anytime after screen 3.

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4.3.1 Power burn-in conditions. Burn-in conditions are as follows: MIL-STD-750 method 1038, test condition A. T_C = + 125°C; V_R = 80 percent of rated V_{RWM} dc.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements) The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit and conditions for $Z_{\theta JX}$ in screening (appendix E, table IV of MIL-PRF-19500) shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit (not exceed the group A limit). Once a fixed limit has been established, monitor all future sealing lots using a sample from each lot to be plotted on the applicable SPC chart.

4.3.2.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750. Method 3101 (read and record data $Z_{\theta JX}$) derived conditions, limits and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein. Electrical measurements (end points) shall be as specified in table I subgroup 2 herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measurement current 10 mA.

b. I_H forward heating current 10-20 A.

c. t_M heating time 10 ms.

d. t_{MD} measurement delay time 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 2.7 °C/W per side.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with Table I subgroup 2 herein. Delta requirements shall be in accordance with Table III herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	4066	I_{FSM} = 200 A; 1 surge of 8.3 ms superimposed on I_O. Condition A, T_A = 25°C; V_R = 0; I_O = 10 A continuous half-wave.
B4	1037	Each diode: I_F or I_O = 2 A (minimum) ΔT_J = 85°C, + 15°C, - 5°C for 2,000 cycles minimum.
B5	1038	Condition B, T_{JMAX} = +150°C, T_A = 100°C, 240 hrs.
B6	3101	Limit for thermal resistance is 2.80ºC/W per side.

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4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	4066	I_{FSM} = 200 A; 1 surge of 8.3 ms superimposed on I_O. Condition A, T_A = 25°C; V_R = 0; I_O = 10A continuous half-wave.
B3	1037	Each diode: I_F or I_O = 2 A (minimum) ΔT_J = 85°C, +15°C, - 5°C for 2,000 cycles minimum.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with table I subgroup 2 herein. Delta requirements shall be in accordance with Table III herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C3	2006	Conditions are as follows: X_1, Y_1, Z_1 and Z_2 axis.
C5	3101	$R_{\theta JC}$ = 2.80°C/W maximum, per side.
C6	1037	Each diode: I_F or I_O = 2 A (minimum) ΔT_J = 85°C, +15°C, -5°C for 6,000 cycles minimum.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or requalification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance shall be measured as follows in accordance with method 3101.

a. I_M ---------- 10 mA.

b. I_H ---------- 5 - 10 A.

c. t_{MD} ---------- 100 µs maximum.

d. $R_{\theta JC}$ ---------- 2.80°C/W (maximum per side).

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 3/	3101	See 4.4.1	$Z_{\theta JX}$		2.70	°C/W
Forward voltage	4011	I_{FM} = 3 A (dc) pulsed (see 4.5.1).	V_{F1}		0.62	V dc
		I_{FM} =10 A (dc) pulsed (see 4.5.1).	V_{F2}		0.75	V dc
		I_{FM} =20 A (dc) pulsed (see 4.5.1).	V_{F3}		0.88	V dc
Reverse current leakage	4016	DC method, (see 4.5.1.)	I_{R1}		100	µA dc
1N6840		V_R = 35 V dc				
1N6841		V_R = 45 V dc				
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method, pulsed (see 4.5.1).	I_{R2}		15	mA dc
1N6840		V_R = 35 V dc				
1N6841		V_R = 45 V dc				
Forward voltage	4011	I_F = 10 A pulsed (see 4.5.1)	V_{F4}		0.63	V dc
		I_F = 20 A pulsed (see 4.5.1)	V_{F5}		0.70	
Forward voltage	4011	T_A = -55°C	V_{F6}		0.85	V dc
		Pulsed (see 4.5.1), I_F = 10 A (dc).				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc; f = 1 MHz, V_{SIG} = 50 mV (p-p) (max)	C_J		400	pF
Subgroup 5						
Insulation resistance (Dielectric withstanding voltage)	1016	V_R = 600 V dc; from lid to bottom case. All terminals shorted.	D_{WV}		10	µA
Subgroup 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ All measurements are for each side.
3/ If 4.4.1 test conditions are performed in 100 percent screening, this test need not be performed in group A.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification Inspection
	Method	Conditions	
Subgroup 1			38 devices, c = 0
Thermal shock (temperature cycling)	1051	500 cycles. Condition G	
Hermetic seal	1071		
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			38 devices, c = 0
Burn-in (steady-state blocking life)	1038	t = 1,000 hours, T_J = + 125°C; V_R = 80% of rated V_{RWM}	
Electrical measurements		See table I, subgroup 2.	
Subgroup 3			
Not applicable			
Subgroup 4			5 devices, c = 0
Thermal resistance	3101	See 4.5.2, $R_{\theta JC}$ = 2.80°C/W	
Subgroup 5			5 devices, c = 0
Surge	4066	Condition A, T_A = +25°C I_{FSM} = 200 A, 1 surge of 8.3 ms superimposed on I_O. V_R = 0; I_O = 10 A pk half sine wave, continuous	
Electrical measurements		See table I, subgroup 2.	

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TABLE III. Groups B and C, delta measurement. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 10 A (pk) pulsed (see 4.5.1)	ΔV_{F2}		± 50	mV
2.	Reverse current leakage	4016	DC method $V_R = V_{RWM}$	ΔI_{R1}		25 µA or 100 percent of initial reading, which ever is greater.	

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
- a. Subgroup 3, see table III herein, steps 1 and 2.
- b. Subgroup 4, see table III herein, steps 1 and 2.
- c. Subgroup 5, see table III herein, steps 1 and 2.

2/ The electrical measurements for table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows:
- a. Subgroup 2, see table III herein, steps 1 and 2.
- b. Subgroup 3, see table III herein, steps 1 and 2.
- c. Subgroup 6, see table III herein, steps 1 and 2.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
- a. Subgroups 2 and 3, see table III herein, steps 1 and 2 for all levels.
- b. Subgroup 6, see table III herein, steps 1 and 2 for all levels.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

d. Type designation and quality assurance level.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

Custodians:
Navy - EC
NASA - NA
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/678	2. DOCUMENT DATE 000701

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, DIODE, SILICON, DUAL SCHOTTKY CENTER TAP POWER RECTIFIER, SURFACE MOUNTED TYPES 1N6840U3 AND 1N6841U3 JAN, JANTX, JANTXV AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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